

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen				Direct Printing Legend	GBR
F.Mask		0,02mm		Solder Resist	Solder Mask GBR
Copper	L1 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,0152mm (0,43oz)		Plane	GBR
Core		0,5579mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,0152mm (0,43oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Core		0,5579mm	FR4	Dielectric	
Copper	L5 (GND)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4	Dielectric	
Copper	L6 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
B.Mask		0,02mm		Solder Resist	Solder Mask GBR
B,Silkscreen				Direct Printing Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,6262mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

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FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY PALROC LABS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE102.500 x 90.000 mm
BOARD THICKNESS1.626 mm
TRACE WIDTH0.152 mm
TRACE TO TRACE0.150 mm
MIN. HOLE (PTH)0.150 mm
MIN. HOLE (NPTH)0.600 mm
ANNULAR RING0.100 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm

Top Fabrication (Scale 1:1)

- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 604a55f
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 12

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Bottom Fabrication (Scale 1:1)

3x 120°

30°

20.50

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Bottom Fabrication (Scale 1:1)

3x 120°

30°

20.50

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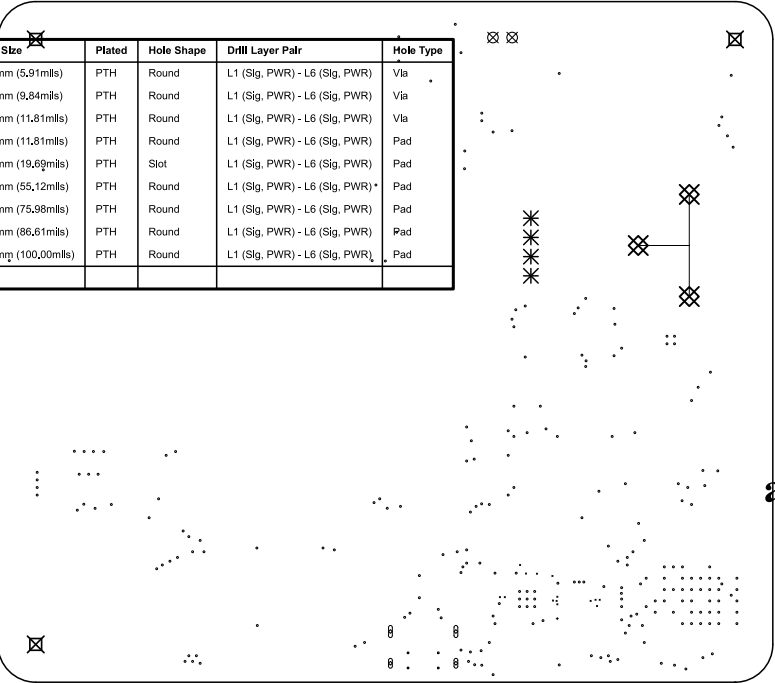
Bottom Fabrication (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	14	0,15mm (5,91mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
○	234	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
+	1	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
□	4	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	4	0,50mm (19,69mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊗	2	1,40mm (55,12mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)*	Pad
※	4	1,93mm (75,98mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊗	3	2,20mm (86,61mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊗	3	2,54mm (100,00mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
Total 269						



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Drill Drawing L1 - L6 (Scale 1:1)

	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 604a55f
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Drill Drawing (L1 - L6)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 12

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0.60mm (23.62mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
○	3	0.99mm (39.00mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
+	4	2.37mm (93.50mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
Total 9						

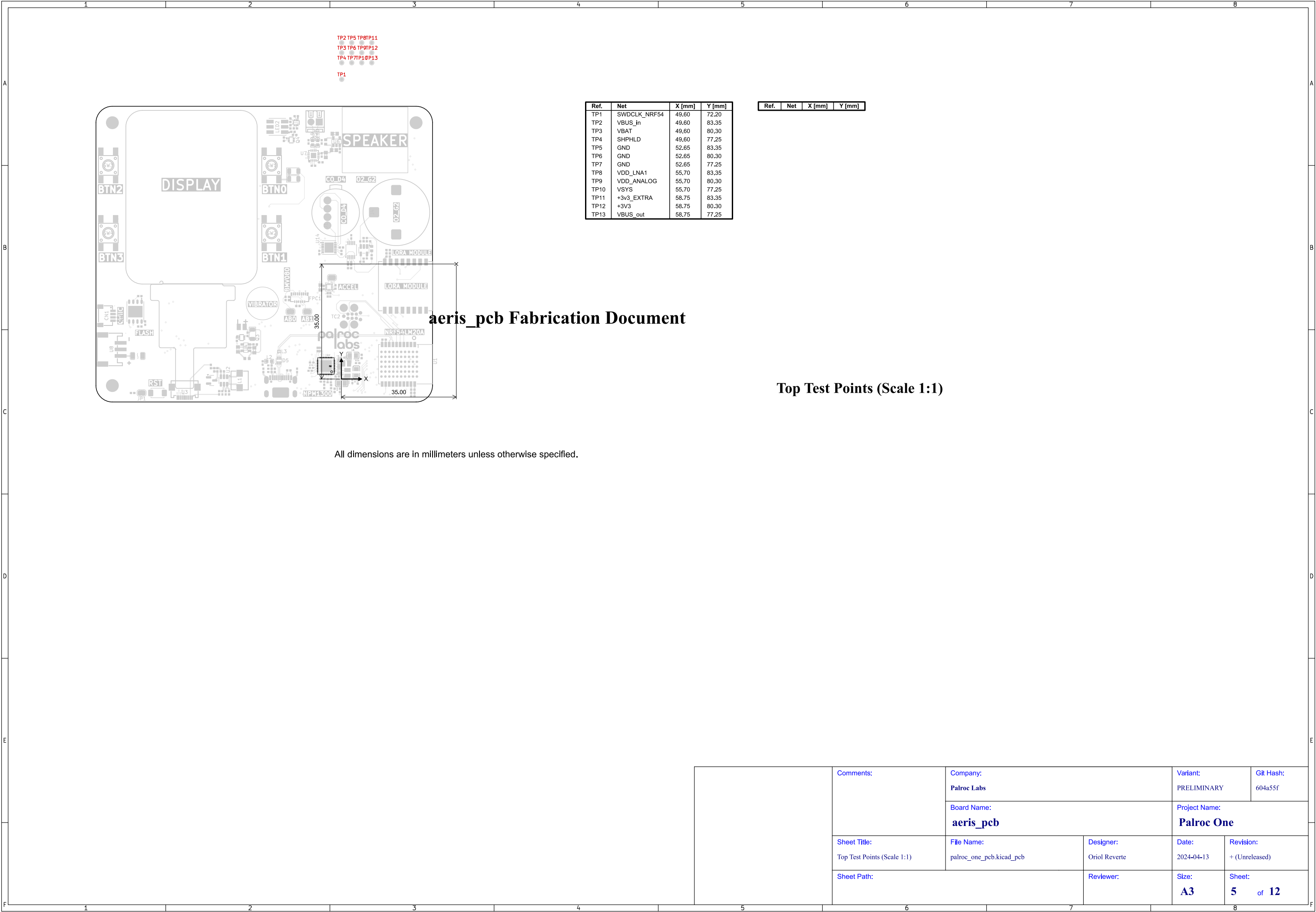


x x

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Drill Drawing L1 - L6 (Scale 1:1)

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		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Drill Drawing (L1 - L6)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 12



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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
1	1	10.00	10.00
2	2	10.00	20.00
3	3	10.00	30.00
4	4	10.00	40.00
5	5	10.00	50.00
6	6	10.00	60.00
7	7	10.00	70.00
8	8	10.00	80.00
9	9	10.00	90.00
10	10	10.00	100.00
11	11	10.00	110.00
12	12	10.00	120.00
13	13	10.00	130.00
14	14	10.00	140.00
15	15	10.00	150.00
16	16	10.00	160.00
17	17	10.00	170.00
18	18	10.00	180.00
19	19	10.00	190.00
20	20	10.00	200.00
21	21	10.00	210.00
22	22	10.00	220.00
23	23	10.00	230.00
24	24	10.00	240.00
25	25	10.00	250.00
26	26	10.00	260.00
27	27	10.00	270.00
28	28	10.00	280.00
29	29	10.00	290.00
30	30	10.00	300.00
31	31	10.00	310.00
32	32	10.00	320.00
33	33	10.00	330.00
34	34	10.00	340.00
35	35	10.00	350.00
36	36	10.00	360.00
37	37	10.00	370.00
38	38	10.00	380.00
39	39	10.00	390.00
40	40	10.00	400.00
41	41	10.00	410.00
42	42	10.00	420.00
43	43	10.00	430.00
44	44	10.00	440.00
45	45	10.00	450.00
46	46	10.00	460.00
47	47	10.00	470.00
48	48	10.00	480.00
49	49	10.00	490.00
50	50	10.00	500.00
51	51	10.00	510.00
52	52	10.00	520.00
53	53	10.00	530.00
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68	68	10.00	680.00
69	69	10.00	690.00
70	70	10.00	700.00
71	71	10.00	710.00
72	72	10.00	720.00
73	73	10.00	730.00
74	74	10.00	740.00
75	75	10.00	750.00
76	76	10.00	760.00
77	77	10.00	770.00
78	78	10.00	780.00
79	79	10.00	790.00
80	80	10.00	800.00
81	81	10.00	810.00
82	82	10.00	820.00
83	83	10.00	830.00
84	84	10.00	840.00
85	85	10.00	850.00
86	86	10.00	860.00
87	87	10.00	870.00
88	88	10.00	880.00
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91	91	10.00	910.00
92	92	10.00	920.00
93	93	10.00	930.00
94	94	10.00	940.00
95	95	10.00	950.00
96	96	10.00	960.00
97	97	10.00	970.00
98	98	10.00	980.00
99	99	10.00	990.00
100	100	10.00	1000.00

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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2	2	10.00	20.00
3	3	10.00	30.00
4	4	10.00	40.00
5	5	10.00	50.00
6	6	10.00	60.00
7	7	10.00	70.00
8	8	10.00	80.00
9	9	10.00	90.00
10	10	10.00	100.00
11	11	20.00	10.00
12	12	20.00	20.00
13	13	20.00	30.00
14	14	20.00	40.00
15	15	20.00	50.00
16	16	20.00	60.00
17	17	20.00	70.00
18	18	20.00	80.00
19	19	20.00	90.00
20	20	20.00	100.00
21	21	30.00	10.00
22	22	30.00	20.00
23	23	30.00	30.00
24	24	30.00	40.00
25	25	30.00	50.00
26	26	30.00	60.00
27	27	30.00	70.00
28	28	30.00	80.00
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30	30	30.00	100.00
31	31	40.00	10.00
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35	35	40.00	50.00
36	36	40.00	60.00
37	37	40.00	70.00
38	38	40.00	80.00
39	39	40.00	90.00
40	40	40.00	100.00
41	41	50.00	10.00
42	42	50.00	20.00
43	43	50.00	30.00
44	44	50.00	40.00
45	45	50.00	50.00
46	46	50.00	60.00
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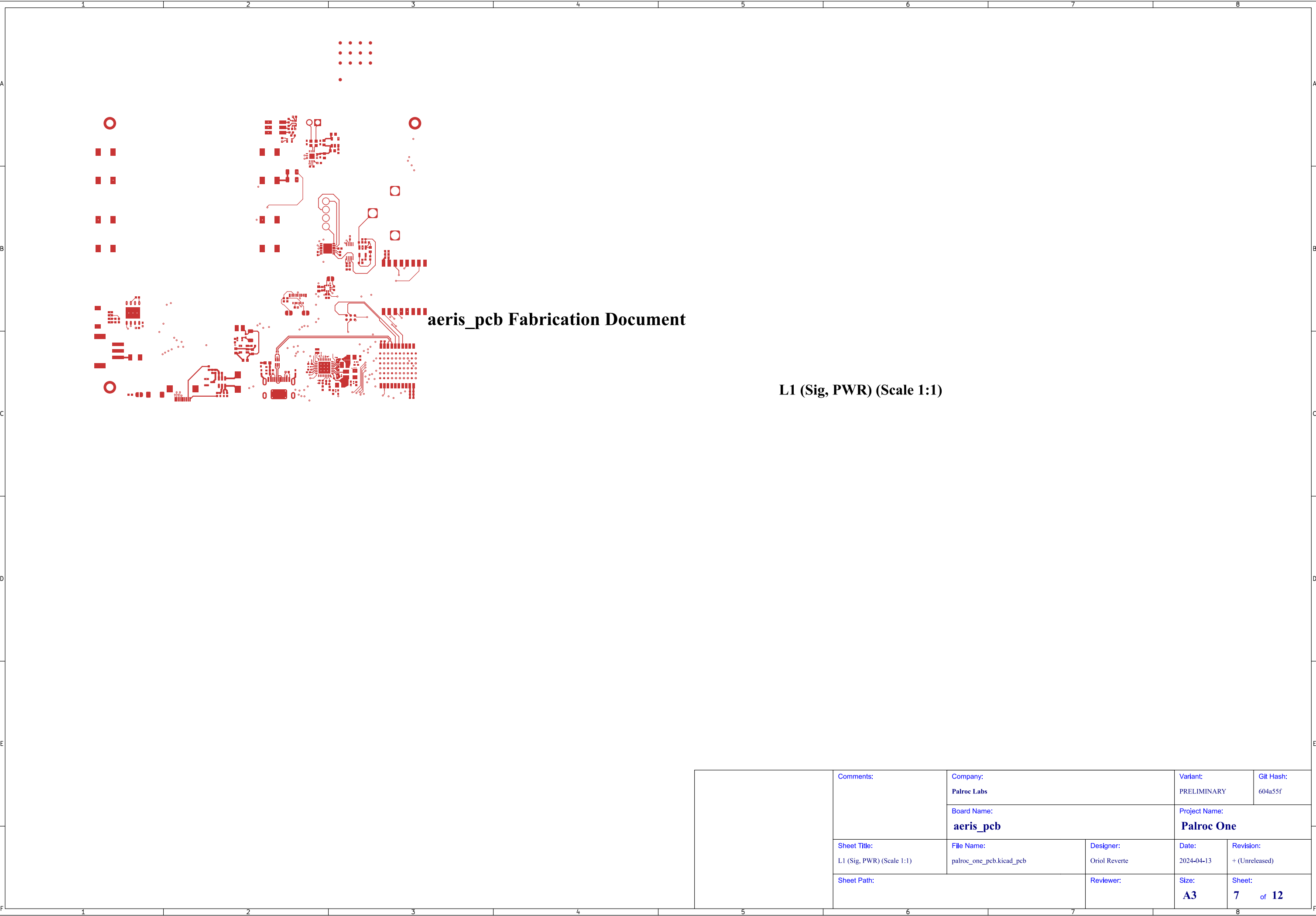
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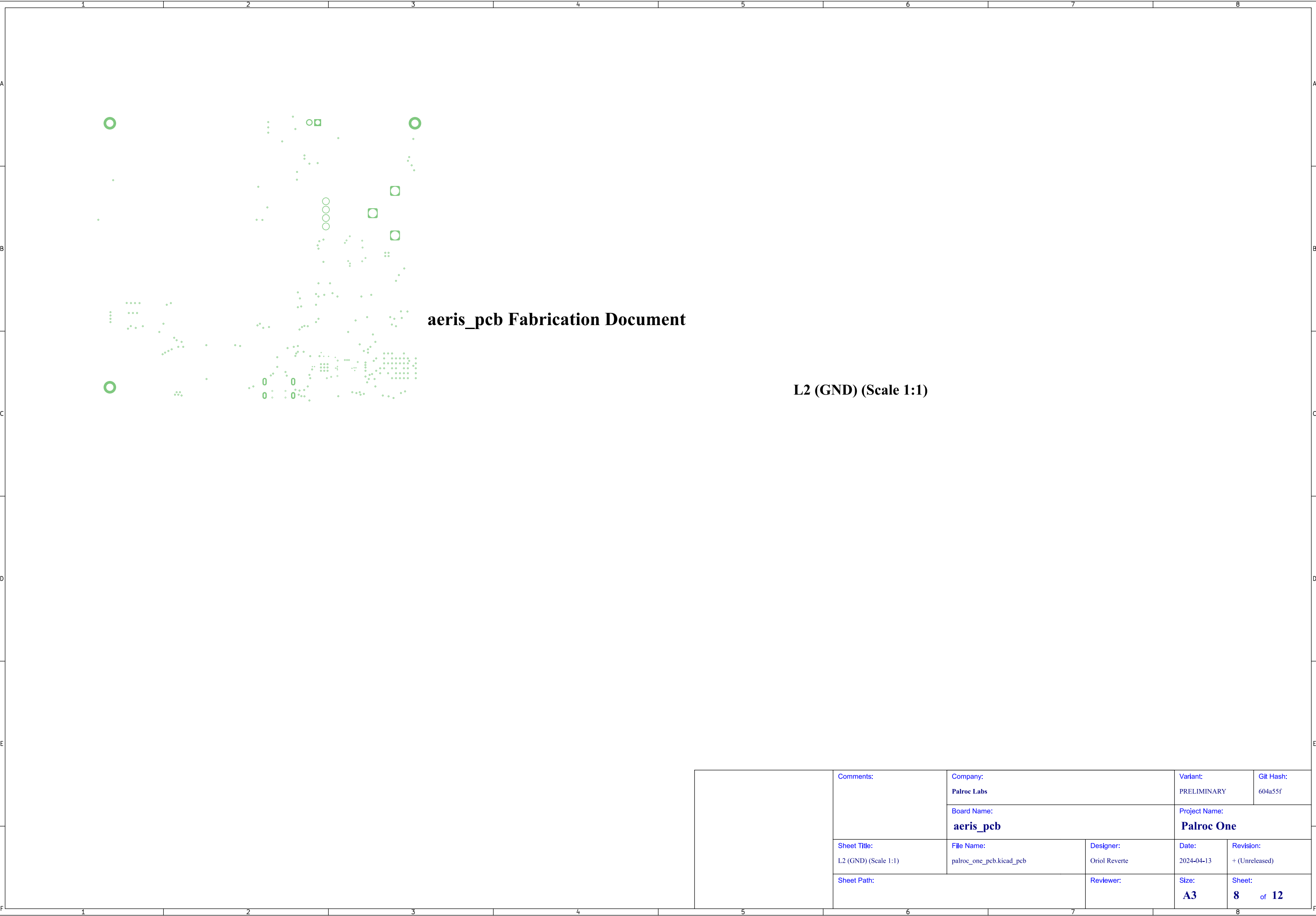
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4	4	10.00	40.00
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10	10	10.00	100.00
11	11	20.00	10.00
12	12	20.00	20.00
13	13	20.00	30.00
14	14	20.00	40.00
15	15	20.00	50.00
16	16	20.00	60.00
17	17	20.00	70.00
18	18	20.00	80.00
19	19	20.00	90.00
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25	25	30.00	50.00
26	26	30.00	60.00
27	27	30.00	70.00
28	28	30.00	80.00
29	29	30.00	90.00
30	30	30.00	100.00
31	31	40.00	10.00
32	32	40.00	20.00
33	33	40.00	30.00
34	34	40.00	40.00
35	35	40.00	50.00
36	36	40.00	60.00
37	37	40.00	70.00
38	38	40.00	80.00
39	39	40.00	90.00
40	40	40.00	100.00

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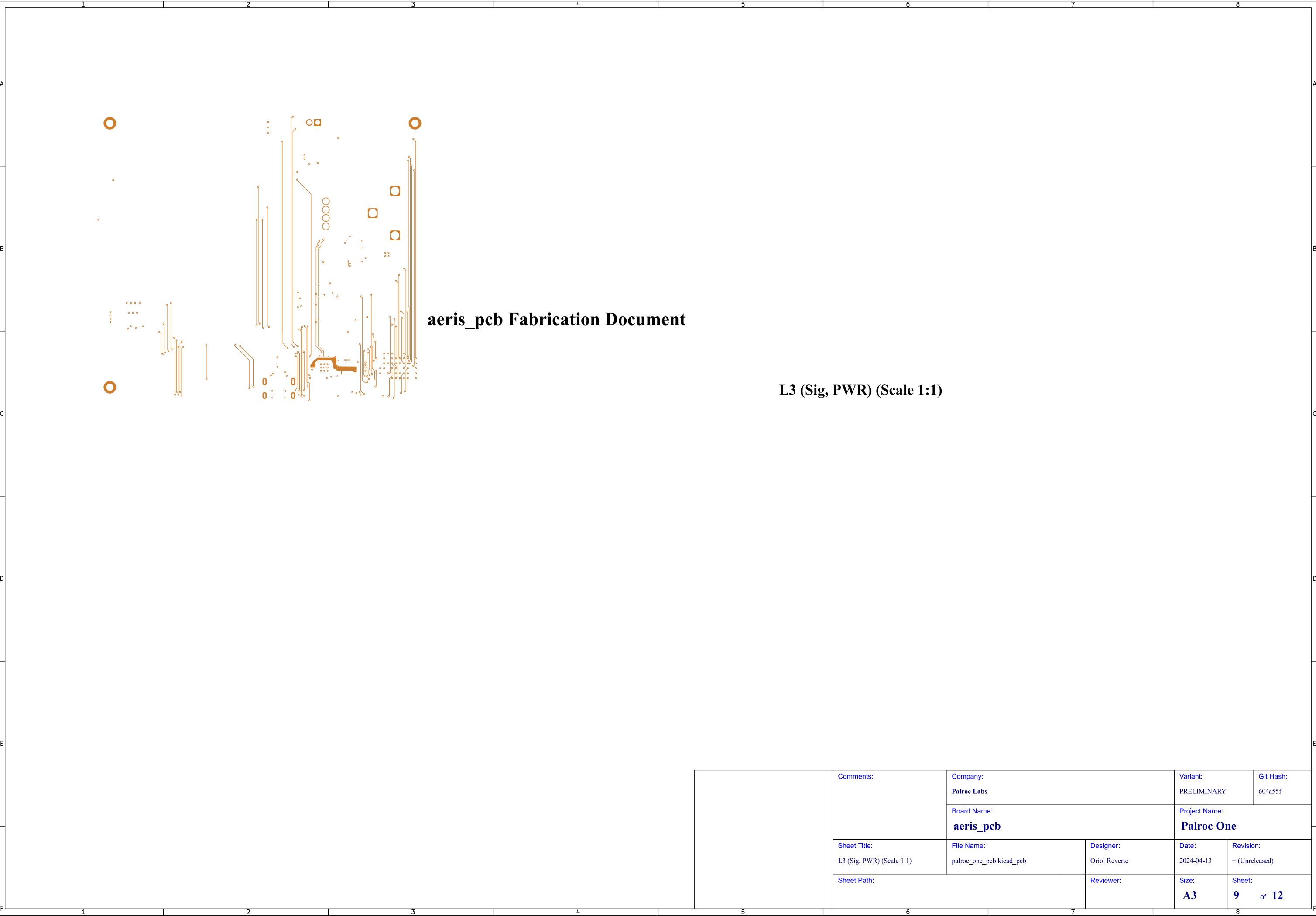




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L2 (GND) (Scale 1:1)

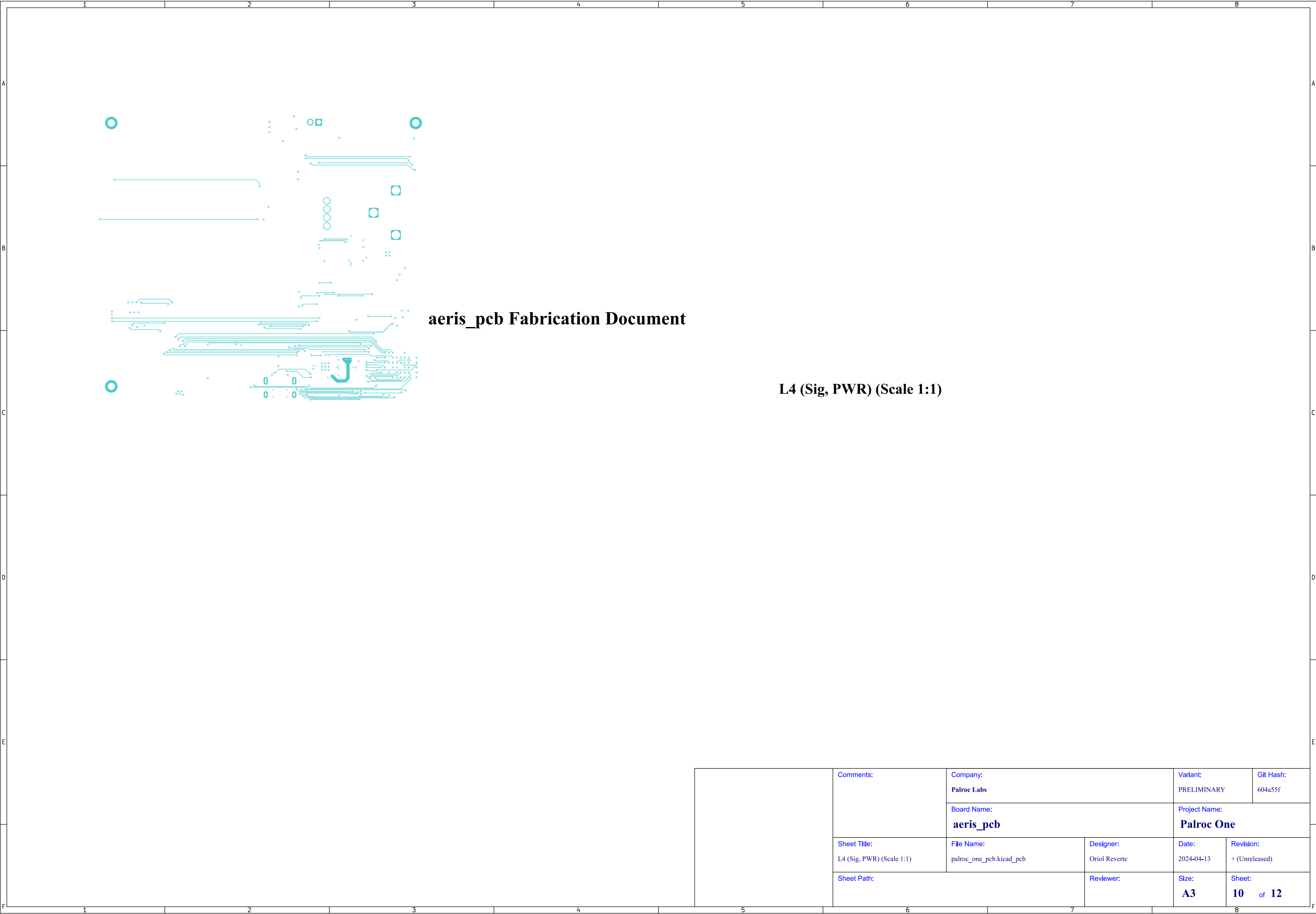
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		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 12



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L3 (Sig, PWR) (Scale 1:1)

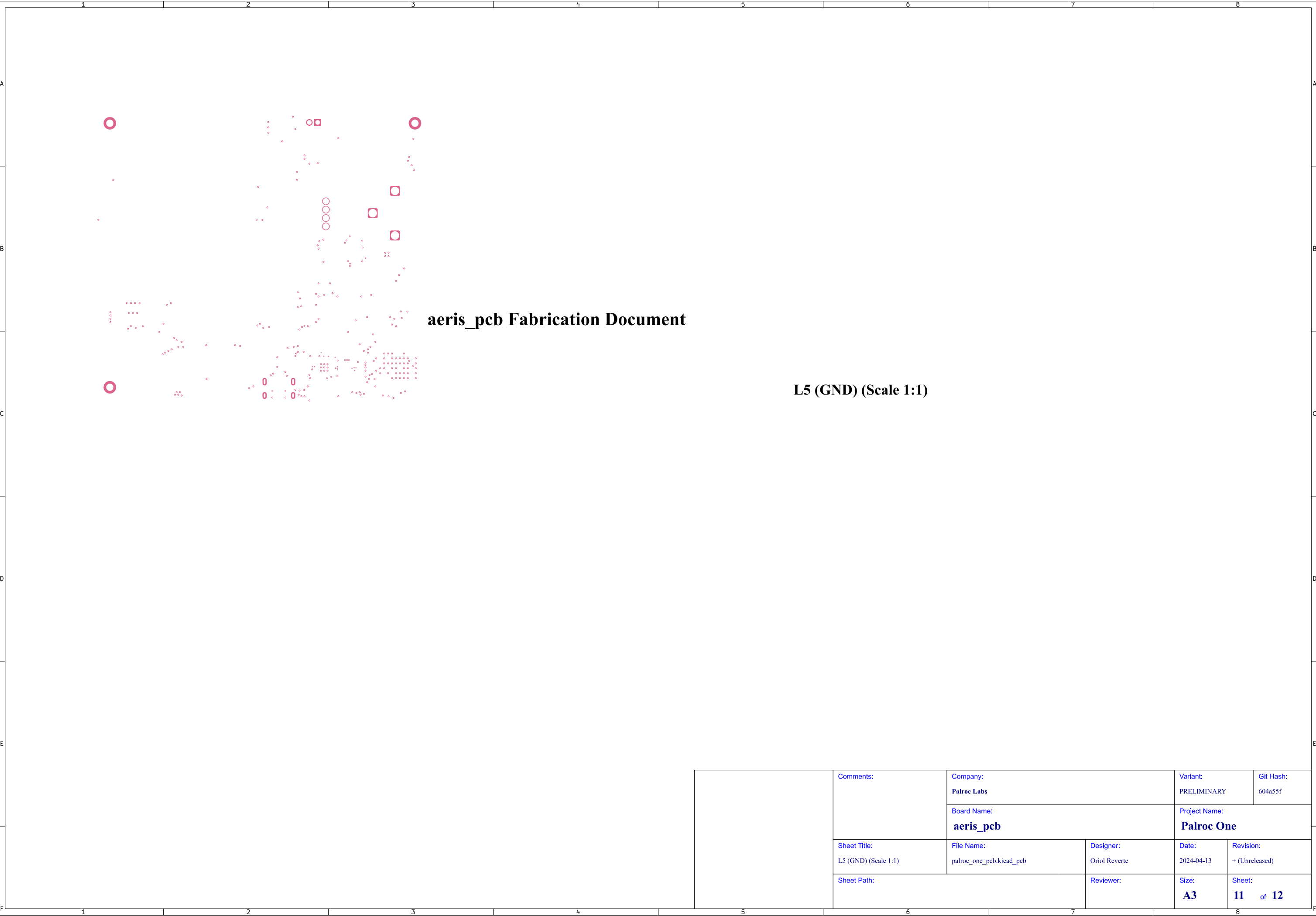
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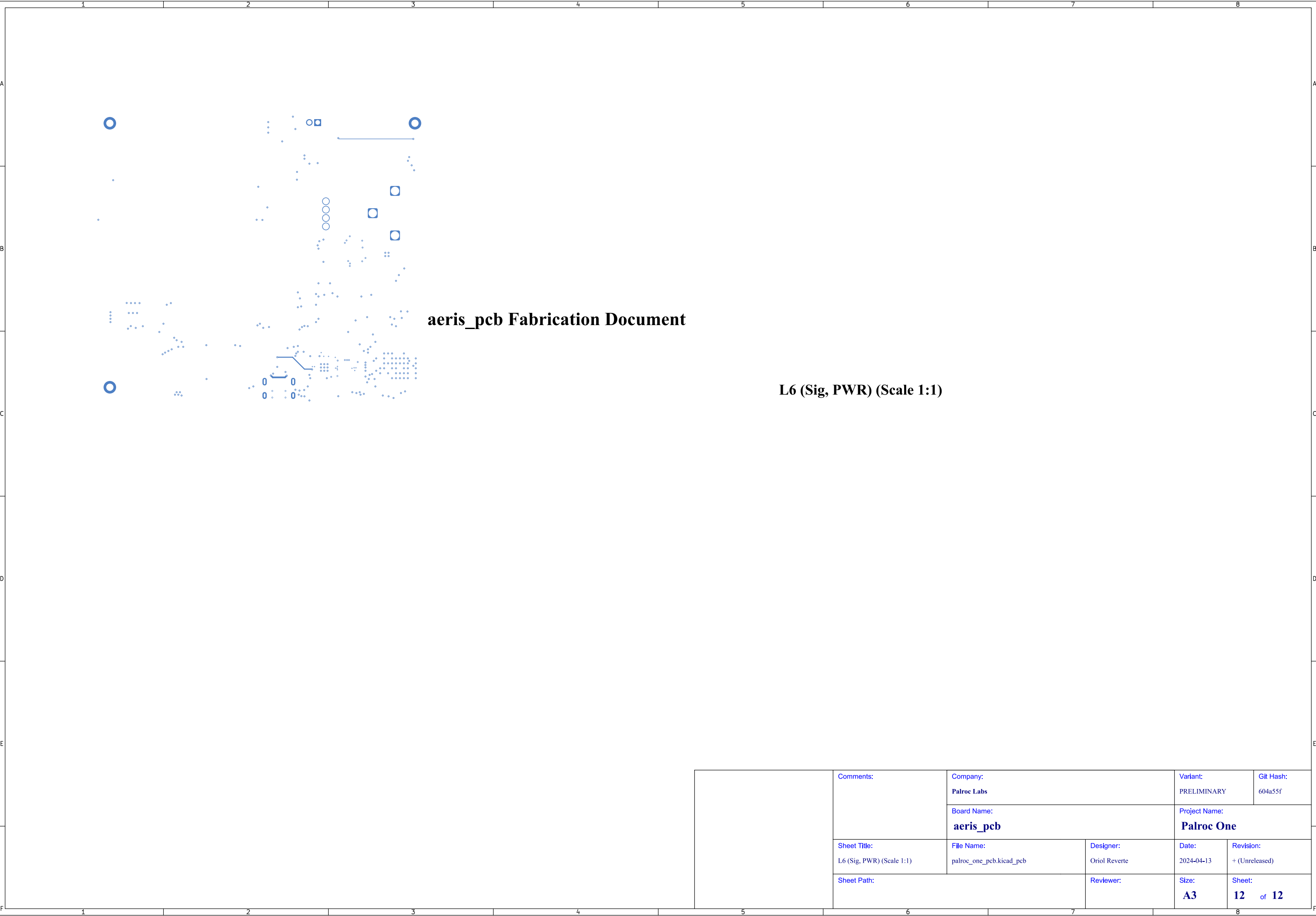
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L4 (Sig, PWR) (Scale 1:1)

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	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 12



	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 604a55f
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: L5 (GND) (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 11 of 12



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L6 (Sig, PWR) (Scale 1:1)

	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 604a55f
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: L6 (Sig, PWR) (Scale 1:1)	File Name: palroc_one_pcb.kicad_pcb	Designer: Oriol Reverte	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 12 of 12